



SUPERSTORE MARKETING CAMPAIGN ANALYSIS

**CUSTOMER SEGMENTATION & GOLD MEMBERSHIP
RESPONSE INSIGHTS**

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BUSINESS CONTEXT & OBJECTIVES

BUSINESS CONTEXT

The Superstore chain launched a Gold Membership campaign offering customers a 20% discount on purchases. The business wants to understand which customers are most likely to accept the membership offer and how customer behavior influences campaign performance.

PROJECT OBJECTIVES

- Identify customer segments with the highest membership acceptance
- Analyze purchasing behavior across different product categories
- Understand the impact of demographics and purchase channels on customer response
- Detect high-value and at-risk customer segments
- Recommend strategies to improve campaign effectiveness and customer retention

KEY BUSINESS QUESTIONS

**WHICH CUSTOMERS
ARE MORE LIKELY TO
ACCEPT THE GOLD
MEMBERSHIP?**

**WHICH PRODUCT
CATEGORIES GENERATE
THE HIGHEST
REVENUE?**

**WHICH PURCHASING
CHANNELS DRIVE
HIGHER ENGAGEMENT?**

**WHICH CUSTOMER
SEGMENTS ARE
UNDERPERFORMING?**

**HOW CAN MARKETING
SPEND BE OPTIMIZED?**

DATASET OVERVIEW & METHODOLOGY

CUSTOMER DEMOGRAPHICS

- Age
- Education
- Marital Status
- Income
- Household Composition

SPENDING VARIABLES

- Wine Products
- Meat Products
- Fish Products
- Fruits
- Sweet Products
- Gold Products

PURCHASE CHANNELS

- Web Purchases
- Store Purchases
- Catalog Purchases
- Discount Purchases

CAMPAIGN VARIABLES

- Response
- Recency
- Complaint Status

TOOLS & TECHNOLOGIES

- SQL for data validation and querying
- Python for cleaning, EDA, feature engineering, and statistical analysis
- Power BI for dashboard development and visualization

DATA QUALITY ACTIONS

- Missing income values were imputed using median values
- Missing spending values were replaced with zero where appropriate
- Unrealistic birth years and age outliers were identified and removed
- Categorical variables were standardized
- Duplicate records were checked and removed

EXECUTIVE KPI SUMMARY

₹1.36M

TOTAL REVENUE

Strong overall campaign performance with significant revenue generation

14.93%

CAMPAIGN RESPONSE RATE

Relatively low response indicating opportunities for targeted marketing

₹606

AVERAGE CUSTOMER SPEND

High-spending customers contributed disproportionately to total revenue

2,000+

TOTAL CUSTOMERS

Digital purchasing behavior strongly influenced campaign acceptance



CUSTOMER DEMOGRAPHIC INSIGHTS



AGE DISTRIBUTION

Most customers fall between the age group of 45–70. Middle-aged customers represent the largest and most active customer base. Younger customers are underrepresented in the dataset.



EDUCATION LEVELS

Graduation-level customers form the largest customer segment. Higher-educated customers contribute significantly to total spending. Customers with advanced education levels show stronger engagement.



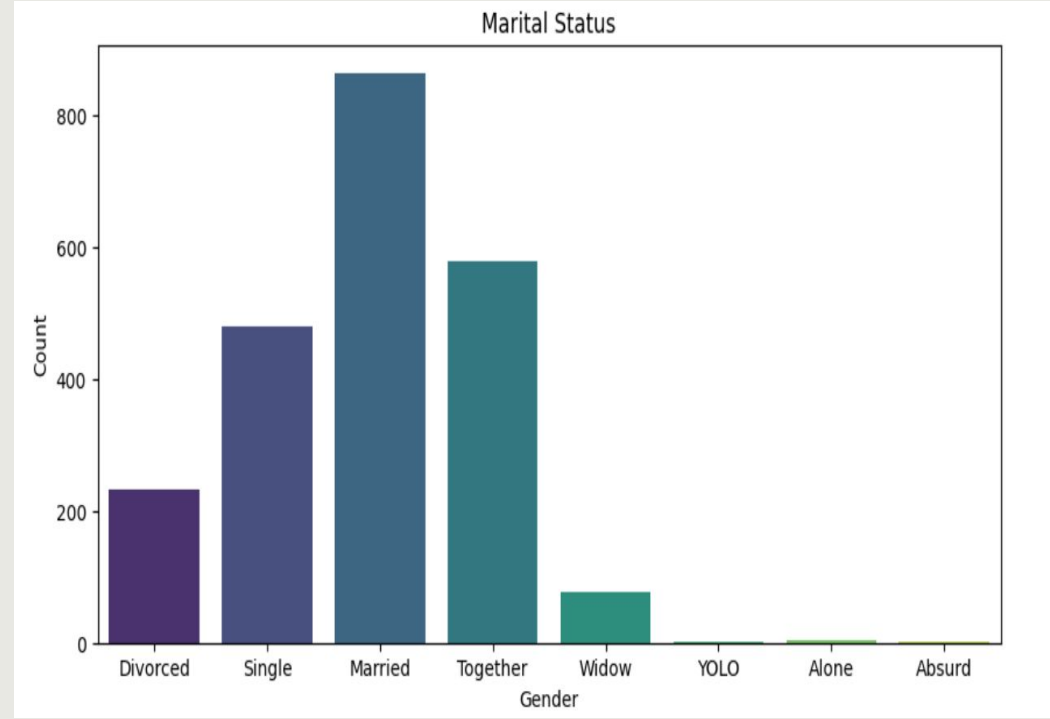
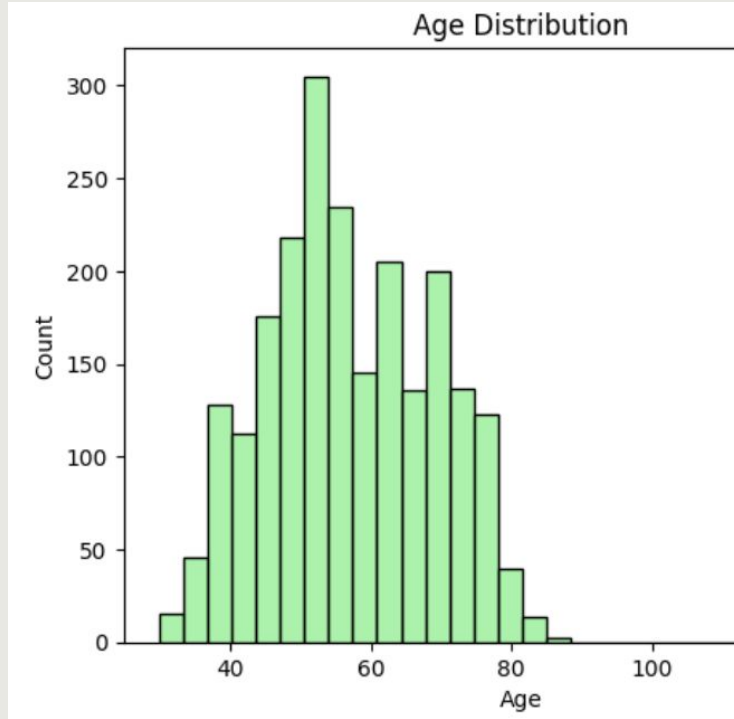
MARITAL STATUS

Married customers represent the largest customer group. Together and Single segments also contribute significantly. Divorced and Widow segments show lower engagement levels.



Key Observation: Customer demographics strongly influence purchasing behavior and campaign response.

CUSTOMER DEMOGRAPHIC CHARTS



PRODUCT & CHANNEL PERFORMANCE ANALYSIS

PRODUCT CATEGORY INSIGHTS

WINE PRODUCTS

Generated the highest overall revenue

MEAT PRODUCTS

Second-highest contributor to total revenue

FRUITS & SWEET PRODUCTS

Generated comparatively lower revenue

PURCHASE CHANNEL INSIGHTS

STORE PURCHASES

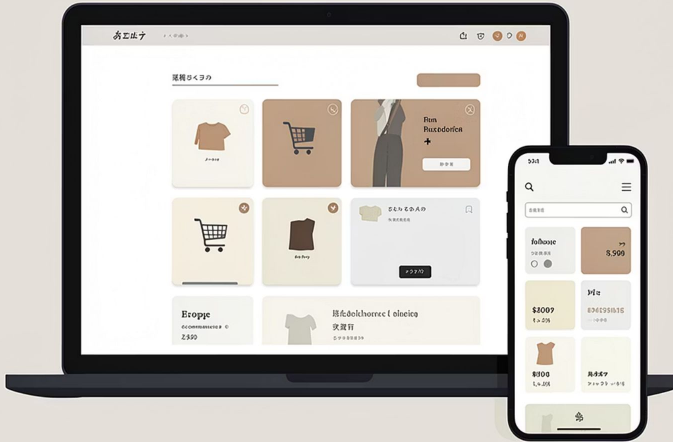
Remain the dominant purchasing channel

WEB PURCHASES

Demonstrate strong engagement and campaign responsiveness

CATALOG PURCHASES

Contribute moderately to total sales



Key Business Observation: Customers who actively use digital channels are more likely to respond positively to the membership campaign.

CUSTOMER SEGMENTATION INSIGHTS



HIGH-VALUE CUSTOMER SEGMENT

High income, high total spending, frequent web purchases, multiple category purchases, and higher campaign response rate.



AT-RISK CUSTOMER SEGMENT

Low spending, high recency, low engagement, limited category purchases, and low campaign response rate.



DIGITAL CUSTOMER SEGMENT

High web activity, frequent online purchases, and higher membership acceptance.



FAMILY-ORIENTED SEGMENT

Households with children or teenagers, but lower response rates compared to other groups.

Key Observation: Different customer segments require different marketing strategies for improved campaign performance.

STATISTICAL & PREDICTIVE ANALYSIS

STATISTICAL FINDINGS

θ_1

T-TEST ANALYSIS

Income levels showed significant differences between responders and non-responders

θ_2

ANOVA ANALYSIS

Customer spending varied significantly across education levels

θ_3

CHI-SQUARE ANALYSIS

Education level and campaign response showed meaningful association

PREDICTIVE MODELING

Machine learning models were developed to predict membership acceptance.

MODELS USED

- Logistic Regression
- Random Forest

KEY PREDICTORS OF CAMPAIGN RESPONSE

- Total Spend
- Income
- Web Purchases
- Customer Engagement
- Recency



Key Conclusion: Customers with higher spending and stronger digital engagement are more likely to accept the Gold Membership offer.

STRATEGIC RECOMMENDATIONS



TARGET HIGH-INCOME CUSTOMERS

Develop personalized premium campaigns and exclusive offers for high-value customers.



EXPAND DIGITAL MARKETING

Increase focus on email marketing, web campaigns, and personalized online offers for digitally active customers.



RE-ENGAGE AT-RISK CUSTOMERS

Offer discounts, loyalty rewards, and personalized promotions to customers with high recency and low spending.



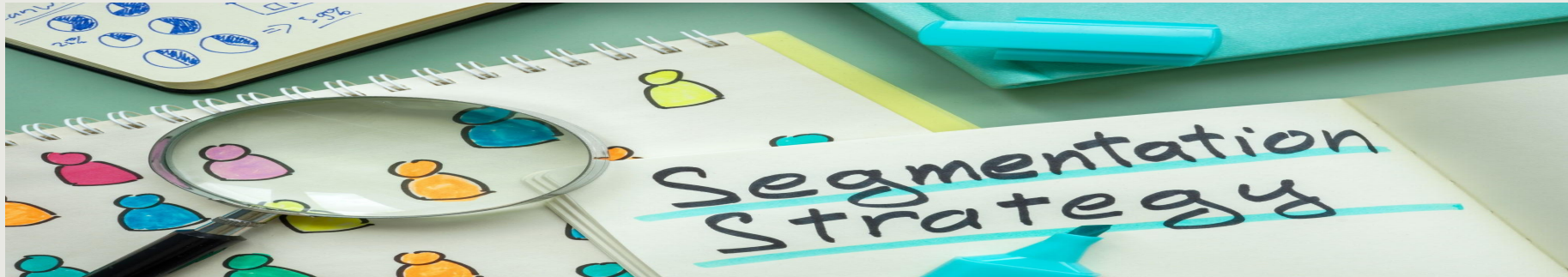
FAMILY-ORIENTED PROMOTIONS

Introduce family bundles and household-focused offers to improve response among customers with children.



CROSS-SELLING OPPORTUNITIES

Promote bundled product offers across multiple categories to increase total customer spend.



EXPECTED BUSINESS

IMPACT

- Higher campaign conversion rates
 - Improved customer retention
 - Increased average spending
 - Better marketing ROI
 - More efficient customer targeting
-

FINAL

CONCLUSION

The analysis revealed that customer income, spending behavior, and digital engagement are the strongest drivers of Gold Membership acceptance. High-value customers demonstrated significantly stronger response rates and contributed the majority of overall revenue. By implementing targeted marketing strategies and leveraging customer behavioral insights, the Superstore chain can improve membership adoption, optimize marketing spend, and strengthen long-term customer relationships.

Thank You

